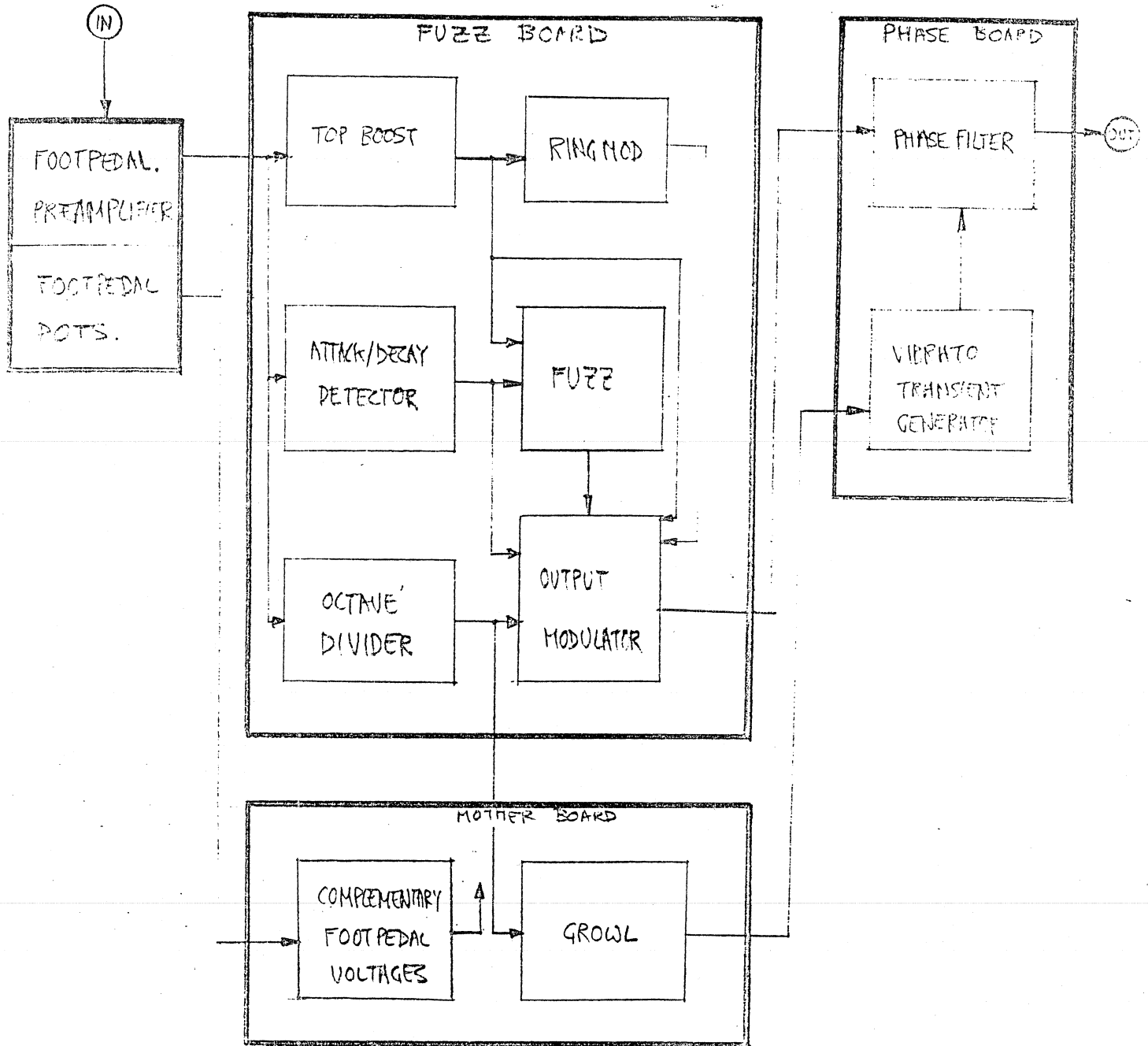


HIFLI
CIRCUITS
AND TEST NOTES

HI FLI EXPLAINED



INTRODUCTION.

The Synthi Hi-Fli combines many well known processes in a single package, such as tone control, fuzz, octave dropping, modulation, phasing, waw-waw, vibrato etc.

The weighting of these effects is ostensibly at the control of the operator and can be manually set and/or controlled by a dual foot pedal and in some cases can be effected by a transient generator which, in turn, is triggered by the operator.

The purpose of this machine is to process musical signals, in particular, guitars.

CIRCUIT DESCRIPTION.

See Drawing No. II3/8.

Foot Pedal.

This is a two-stage, low noise preamplifier. The gain is approximately $\frac{1}{R5}$. Two inputs are provided; one has an

attenuator to make it possible to accept high level inputs, this being necessary because the preamplifier gain never goes less than 3.3 (i.e. 33K/10K). The output of Q2 is AC coupled to a buffer (A1). This is because the buffer drives the rest of the Hi-Fli and it is DC coupled. Any DC offsets would stop many of the circuit functions.

The bypass footswitch provides a -12v or floating signal to the Hi-Fli, the two footpedal pots, merely pick off a control voltage. Their pots, must be aligned to give a symmetrical bipolar output swing.

Fuzz Board.

See Drawing No. II2/6.

The tone burst generator waveforms will be useful to refer to whilst reading the Fuzz and Phase board explanations.

The audio signal from the preamplifier goes to three circuits; the top boost, the octave divider and the attack decay detector.

The Attack Decay Detector. This is used to determine when the input waveform starts and stops. The signal goes into A1 and A2, (I23), where it is equalised and rectified, (see TB 2,3), then it is logarithmically compressed by A6 and A2 (5,6,7) (see TB4). The compression is due to the exponential characteristics of the dual transistor A6 in the feed back loop around A2 (5,6,7). Now the compressed rectified signal is passed through a low pass filter, A3, which is a two stage Sallen Key Filter, cut off slope = -24dB/oct. The result (see TB5) is a compressed image of the input signals envelope.

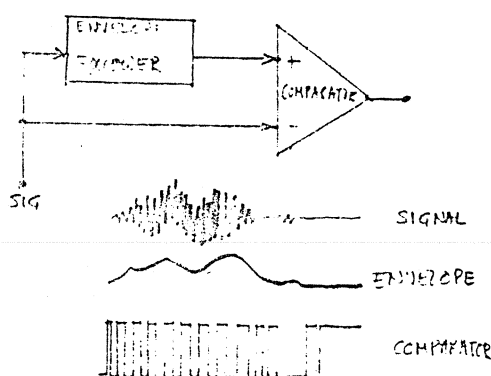
To detect transients in this envelope the waveform is differentiated by C9 and R25; also D5,6 to make sure that these transients are greater than $\pm 600\text{mV}$ (see TB6).

The detected transients are used to drive A4 which comprises two schmitt triggers, one used to generate a pulse on an attack A4, (5,6,7.) (see TB8) and one used to generate a pulse on a decay A4 (I23) (see TB7).

The combination of these two, produces a pulse that goes low in between the attack and decay points A5 (I23) (see TB9) and this

is used to trigger the transient generator on the phase card. Also produced is an exponential decay waveform (see TBI0); when A4, pin 7 is low CIO is discharged via D7. When it is high, D7 is reverse biased, Q1 is switched on and so CIO charges up, via the decay rate pot VR1, to 0v. This voltage is monitored by the Voltage follower A5 (5,6,7) (see TBI0)

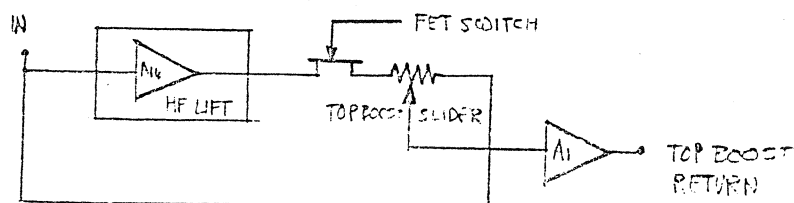
The audio input also goes into an octave divider (A7,8,9.) This circuit is primarily designed to detect the fundamental note of the input and then divided by two. It does this by having complementary envelope followers and complementary comparators.



This arrangement is quite good as a fundamental note detector, and as a complementary system is used as well then a reasonable degree of integrity is to be expected.

The Envelope follower is made up of a perfect rectifier A7, with a fixed time constant. The comparators are A8 (see TB II, I2, I3, I4) which drives A9, a dual D flip-flop. This is a TTL device being powered by +5v which it gets from DI3 R5I. A8 sets and clears the first stage of A9 which produces a neat square wave output (see TBI5). The second half of A9 divides this by two, thus producing a sub octave signal (see TBI6) which is applied to A10. The gain of A10 is controlled by the suboctave slider and the output of A10 is affected by the buzz switch which, when closed, acts as a low pass filter (see TBI7). The output of A10 is used to affect the output modulator to give BUEZ noises.

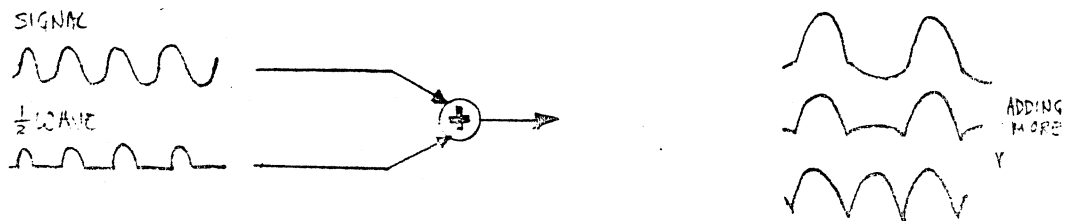
The audio signal also goes into a top boost circuit A14 which gives it a high frequency lift.



The top boost return signal is a mix of the equalised signal and the original, and is fed to the sustain fuzz, ring modulator and output modulator.

AI2 is a peak voltage detector; i.e. it remembers on CI9 the largest positive signal excursion. Therefore, when a note is struck, the peak amplitude is stored on CI9. This signal is fed via the attack rate pot to C20 (see TB20), which is buffered by AI2 (see TB19). The purpose is to make a Fuzz box whose amplitude is proportional to the signal level but whose harmonic structure isn't. The top boost signal is fed into Q6 which is an amplifier whose collector load is driven by the output of AI2. Thus the signal out of AI2 defines the Fuzz modulation envelope (see TB21). The gain of the Fuzz is controlled by AI3, also, when a stop, (end of note), is detected, transistors Q4,5 short out CI9,20 and the Fuzz ceases, ready for the next attack.

Ring Modulation. This is not truly ring modulation but an approximation. Transistors Q2,3 act as a half wave rectifier and AI1 as a controlled modulator, so that as a controlled amount of inverted half wave rectified signal is added to the original the fundamental can be made to completely disappear, (see TB22).



The output modulator is a FET modulator. A matched pair of FET's is used so that the modulator switches off without having need for a preset or having an undesirable gap in the modulation. Also AI6 induces voltages onto the bias FET which in turn controls the modulation FET. The output then goes to the phase board where most of the business occurs.

Phase Board.

See Drawing No. II2/7

This board is by far the most useful in terms of the effects it generates, but although it has less sections than the Fuzz board, it may prove to be more difficult to understand.

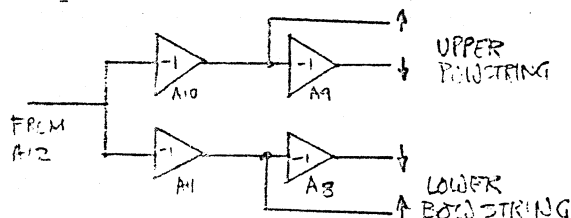
The phase board consists of basically three parts; a function generator, a set of bowstring drivers and a phase filter. The function generator produces ramps, sine waves and attacking and decaying sine waves. The oscillator is a triangle square wave oscillator made from AI6, I5, (see TB23,24).

AI6 is a reversible integrator and AI5 is the schmitt trigger. The speed of oscillation is controlled by the current into pin 5 of AI6, the range switching between $\sim$ and $\sim\sim$ is achieved by transistor Q1 being switched on or off. The triangular output is fed to AI4 where the saturation effect of this IC. bends the $\sim$ into a $\sim\sim$ (see TB25) The gain of AI4 can be controlled so that it can multiply a $\sim$ with a ramp to produce a $\sim\sim$. The output of AI4 then goes to AI3 which is a gain control, and then to AI2 which drives the LED's and the bow string drivers. Also at AI2, the 'direct Shift' is produced; that is, a DC level is added to the function. The output of AI2 goes via diodes D262-5, RI05-II0 and LED's 1 & 2 to the bow string drivers, the purpose of this network being to distort the function (see TB29). The ramp functions are generated by AI7, I3 and Q3. The attack rate is determined by the current into pin 5 of AI7; this I.C. being used to charge up C3, Q3 just being an emitter follower.

The capacitor C3 is reset by AI8 which is controlled by the reset pulse from the Fuzz Board (see TB 27,28) the reset pulse just causing AI8 to drive a large current into or out of C3. The mode of the ramp (i.e. up or down) is controlled by the modulation switch. When the RMS of R2I is held positive by the switch, rising ramps are generated, and when it is negative the mode of AI7, AI8 is reversed and the ramps fall. When the modulation switch is in the $\sim\sim$ & $\sim$ modes, Q3 emitter is held high by D6,7 thus preventing the ramp generator from functioning.

In the $\nearrow \searrow$ modes, D4,5 prevent the oscillator from oscillating. In the $\sim\sim$ modes, both ramp generator and oscillator are functioning.

The bow string drivers drive the 200 Diodes in the phase filter. The distorted output from AI2 is fed into a pair of inverters (see TB29,30,31,32)

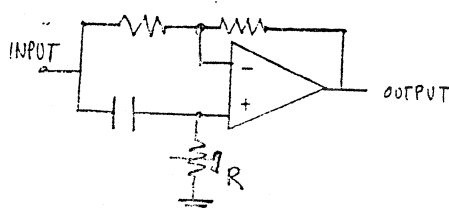


AI0 (I23) adds a DC bias to the drivers which is temperature compensated by D11.

Also AI1 (I23) and Q4 are used as a switchable inverter; when R4I is pulled negative, (MEAN), Q4 is switched on and the lower bow string drivers response to the output of AI2 is inverted.

The bow string drivers drive a string of 20 Diodes, which are forward biased and tapped in the centre. This forms the control for the Phase Filter.

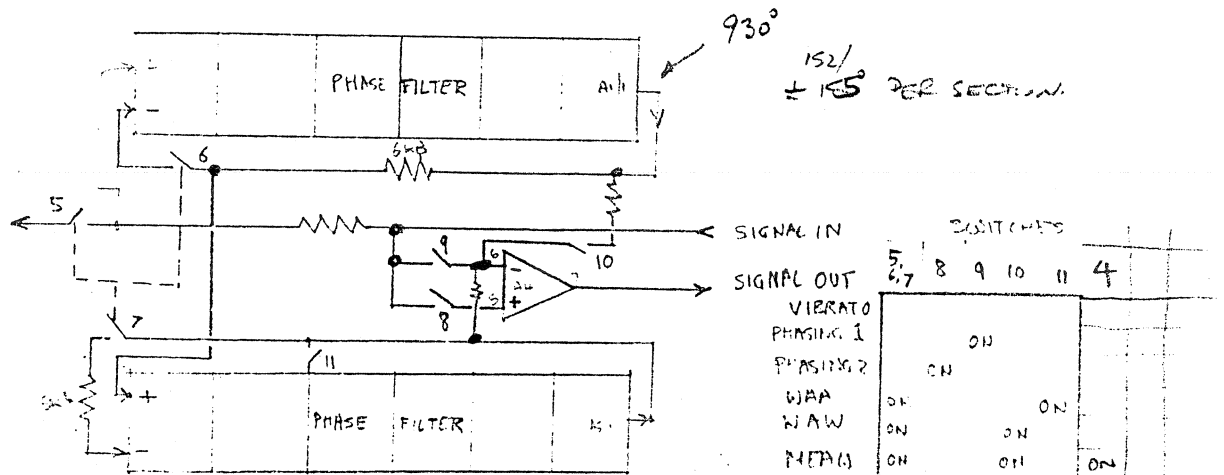
The Phase Filter is a phase delay line, having a delay of up to 12π radians. Each section can give a maximum of π , phase shift.



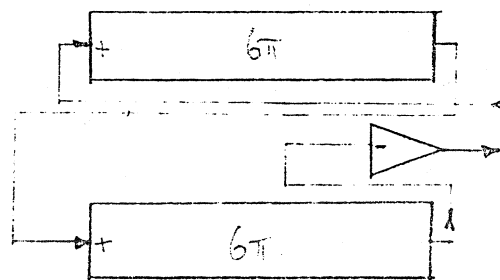
The phase shift circuit is a conventional AI1 pass filter; i.e. a filter that has a fixed amplitude response, but a frequency sensitive phase response.

Alternatively, for a fixed input frequency, the output phase can be varied by varying R. In the case of the Hi-Fli, R is the diode string. Using a controlled phase delay, it is possible to generate vibrato, phasing,

(swept notch) and waa-waw (swept peaks) sounds. To do this we need the services of some routing switches, viz. Q5-II.

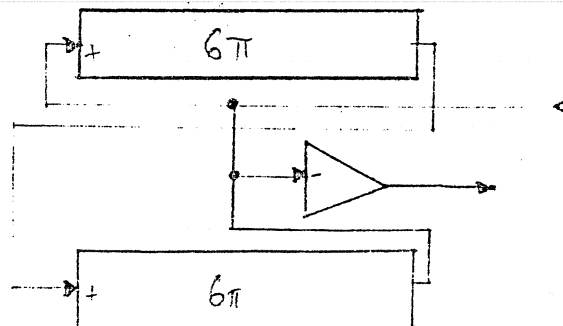


VIBRATO. All switches open.



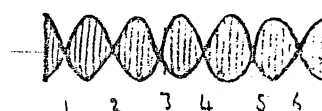
The signal goes through all the phase filters and out the other end. The varying phase shift manifests itself as Vibrato, (i.e. Frequency modulation)

Phasing I.

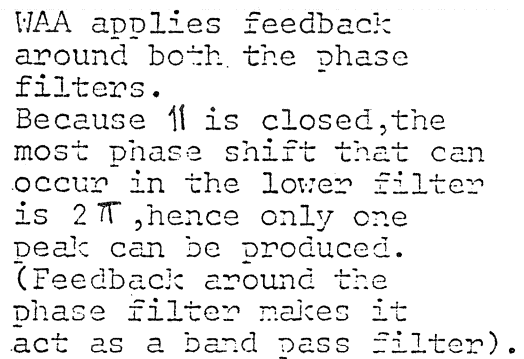


This is almost the same as vibrato except that part of the original signal is added to the phase shifted signal so that cancellation occurs every 2π phase shift.

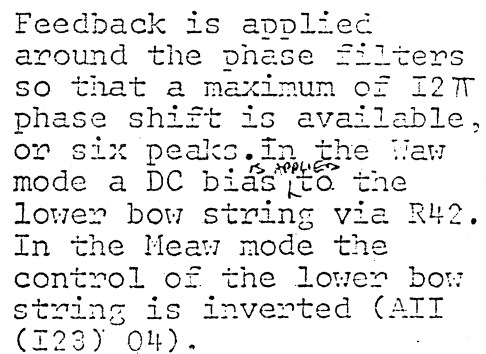
(See wobulator pictures.)



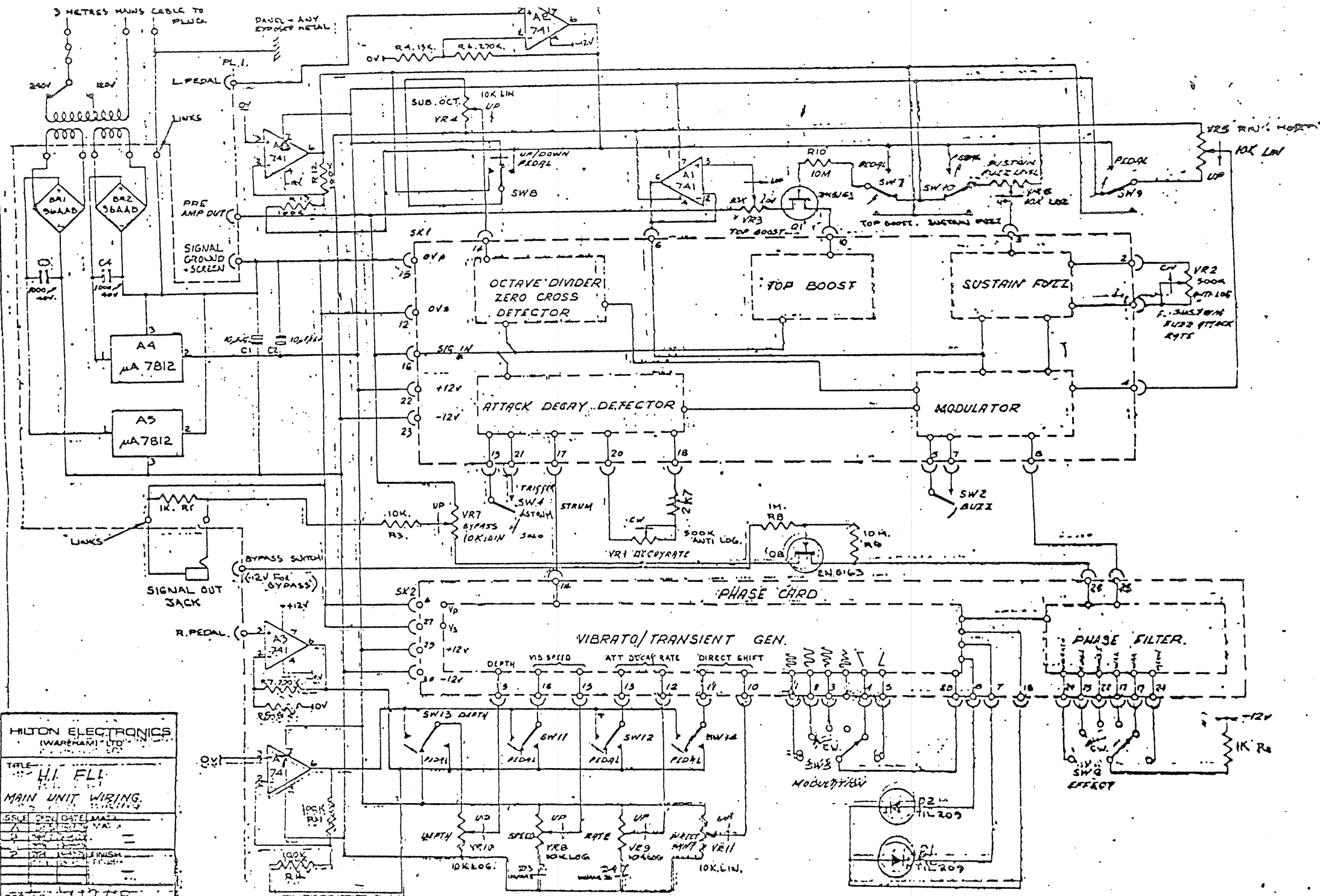
In the three following modes, the audio signal is injected into the bow string drivers.

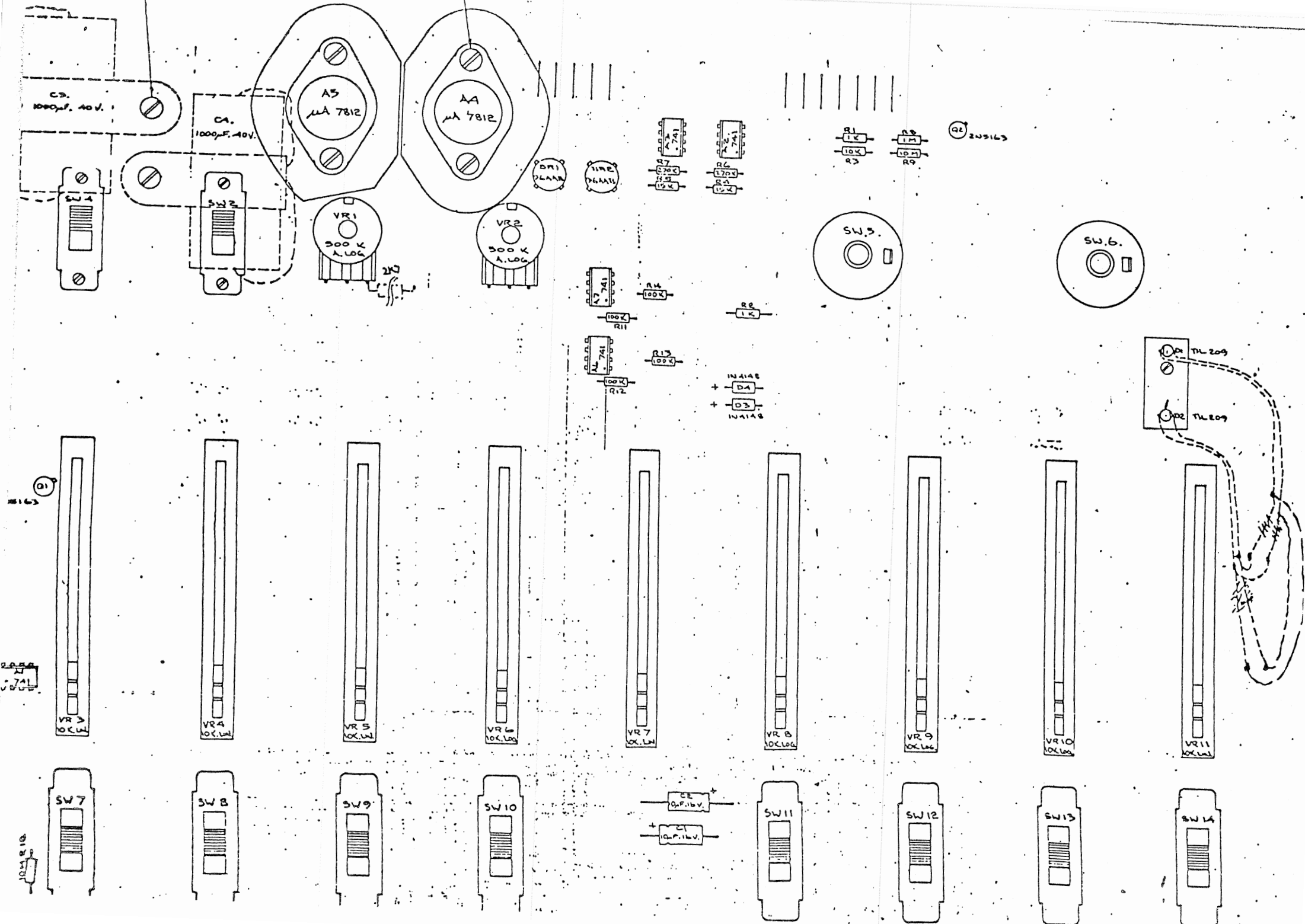


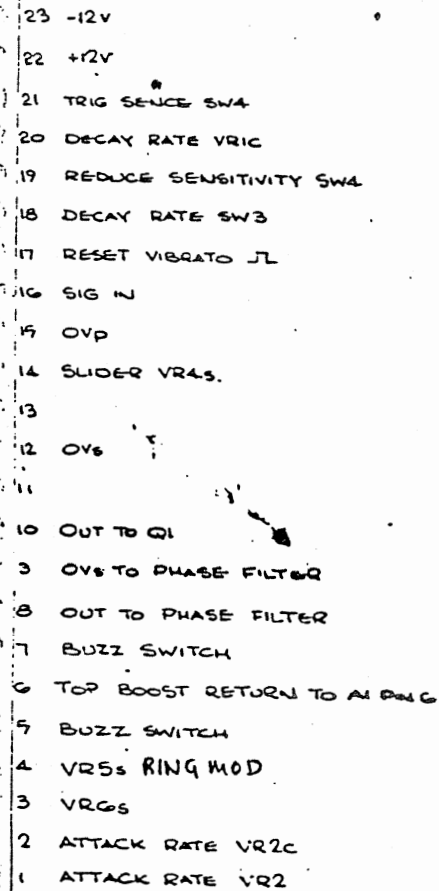
Switches 5,6,7 and I/O are Off for both these modes.



22

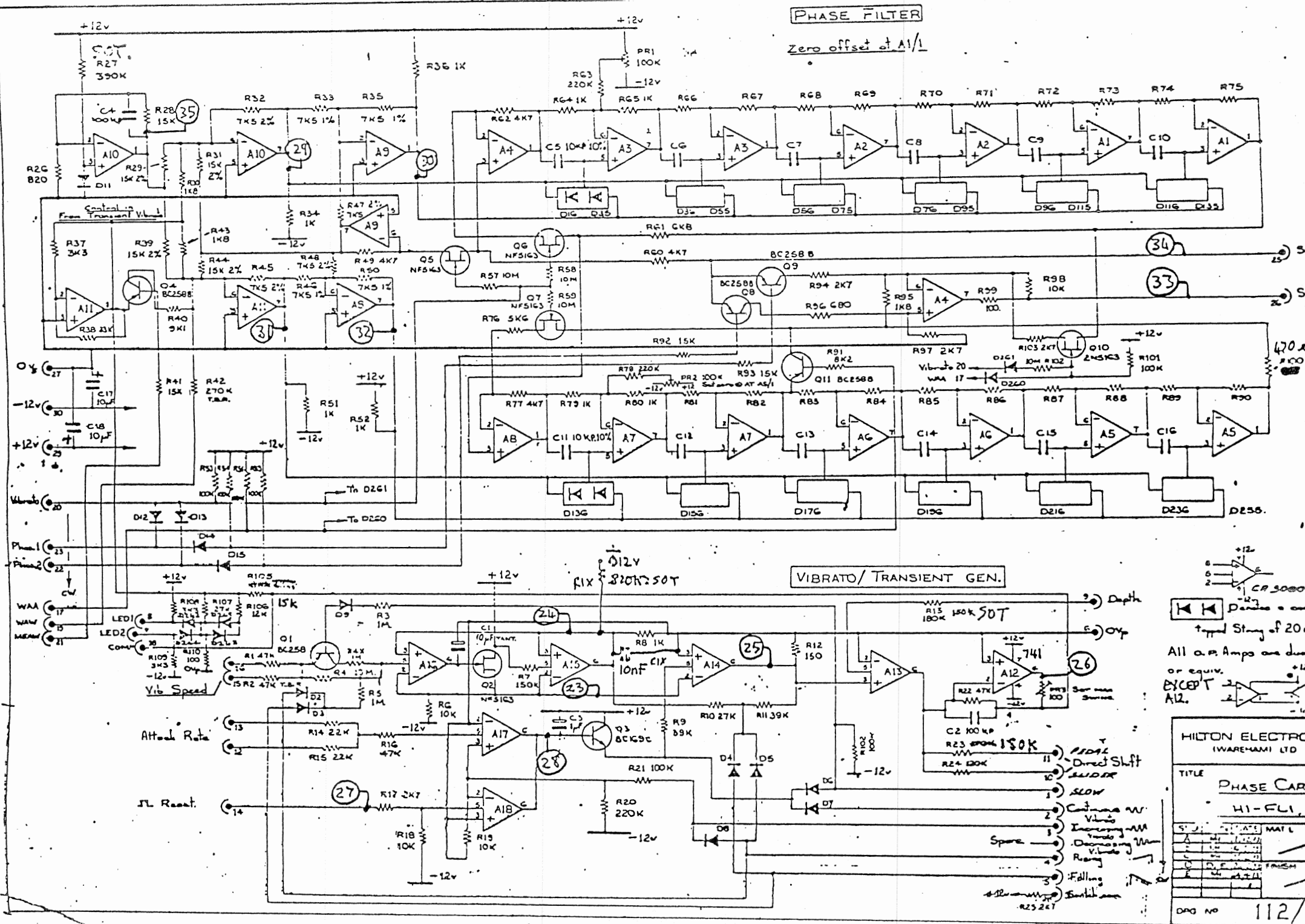




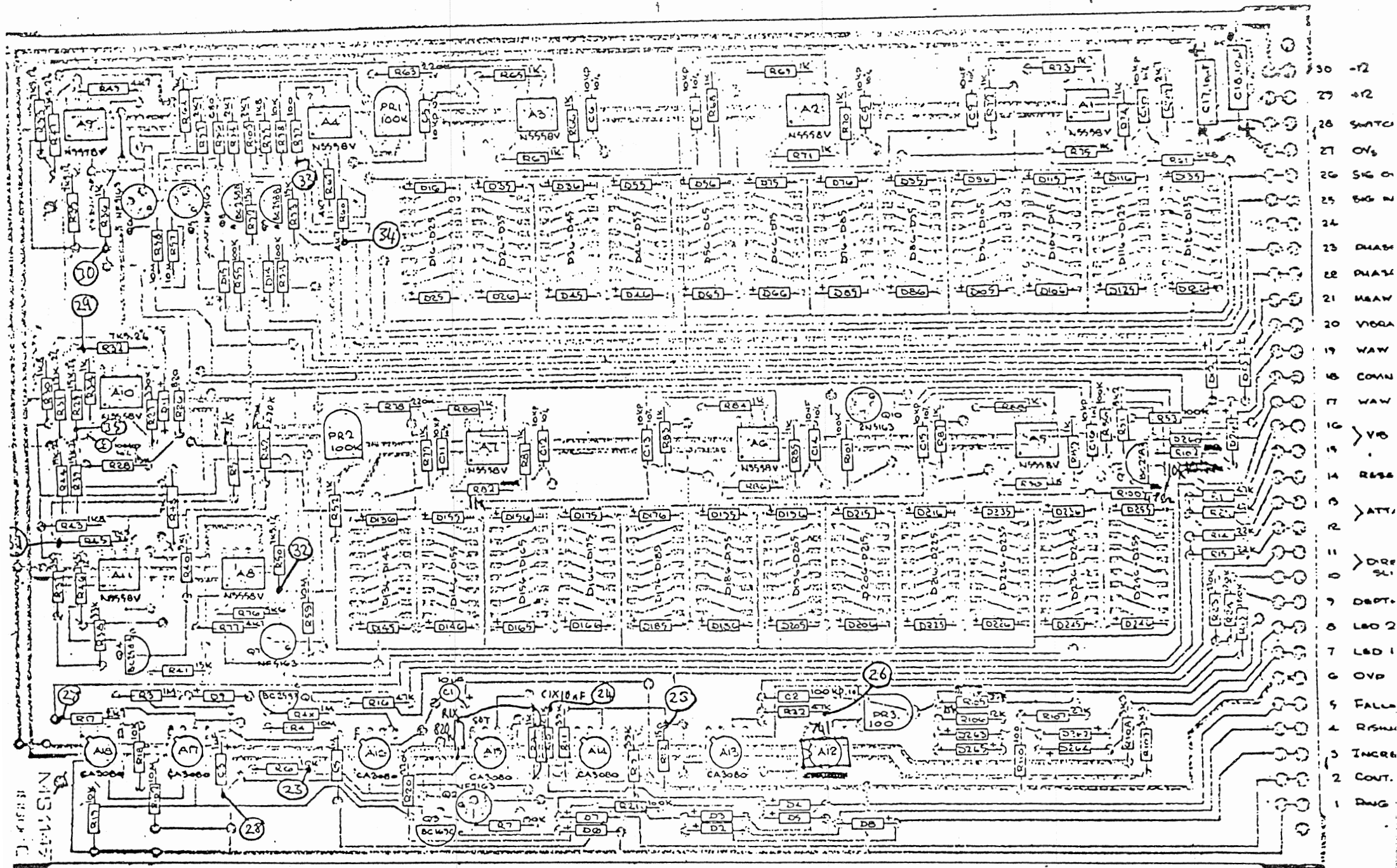


1. ALL DIODES IN4148
2. TANT CAPACITORS C10,
C17, C20 & C26. SPOT TO
FACE P.C.B.
ACCEPT C16, C17.
3. TIS 70% TO BE REALEDITED
TOGETHER BACK TO BACK
BEFORE ASSY.

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200	1/1/71



NOTE:-
 ALL DIODES IN 4140

TEST SCHEDULE FUZZ CARD

USE EXT. TRIGGER
TIME BASE 20ns/cm
REFERENCE Y1

123456 all depend
upon the gain of the
Preamplifier.

③ When the signal
level is increasing
the waveform
suffers from the effects
of diminishing
returns.

The sensitivity
switch sw4,
marked solo, strum,
modifies the Hi-Fi's
sensitivity to
"start" signals.

Turn down signal
level until signal
⑧ nearly disappears;
switch to strum; then
⑨ will disappear.

If decay rate
pot. (ref test 10)
is set at too long
a decay, the
signal size will
become reduced.

①

②③

④

⑤

⑥

⑦

⑧

⑨

⑩

Y2 0V

Y2

Y2

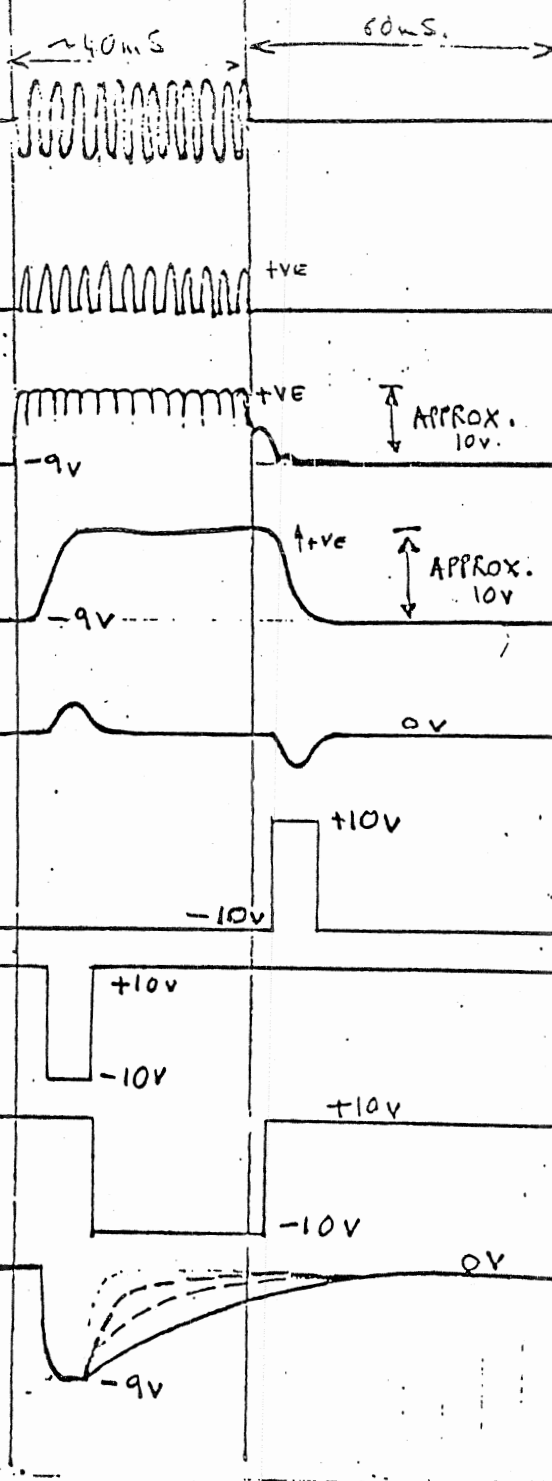
Y2

Y2

Y2

Y2

Y2



SIGNAL IN. PCB: PIN 16

① TONE BURST. 1KHz
Tone burst generator SW1C
TO TONEBURST, FAST.

② junction of D2, R13
" " D4, R14.
1/2 WAVE RECTIFICATION

③ A2 PIN 7
LOGRITHMICALLY COMPRESSED
FULL WAVE RECTIFIED SIGNAL

④ A3 PIN 7.
LOGRITHMIC ENVELOPE
FOLLOWER

⑤ A4 PIN 6.
DIFFERENTIATED ENVELOPE

⑥ A4 PIN 1
HYSTERIC STOP DETECTION

⑦ A4 PIN 7
HYSTERIC START DETECTION

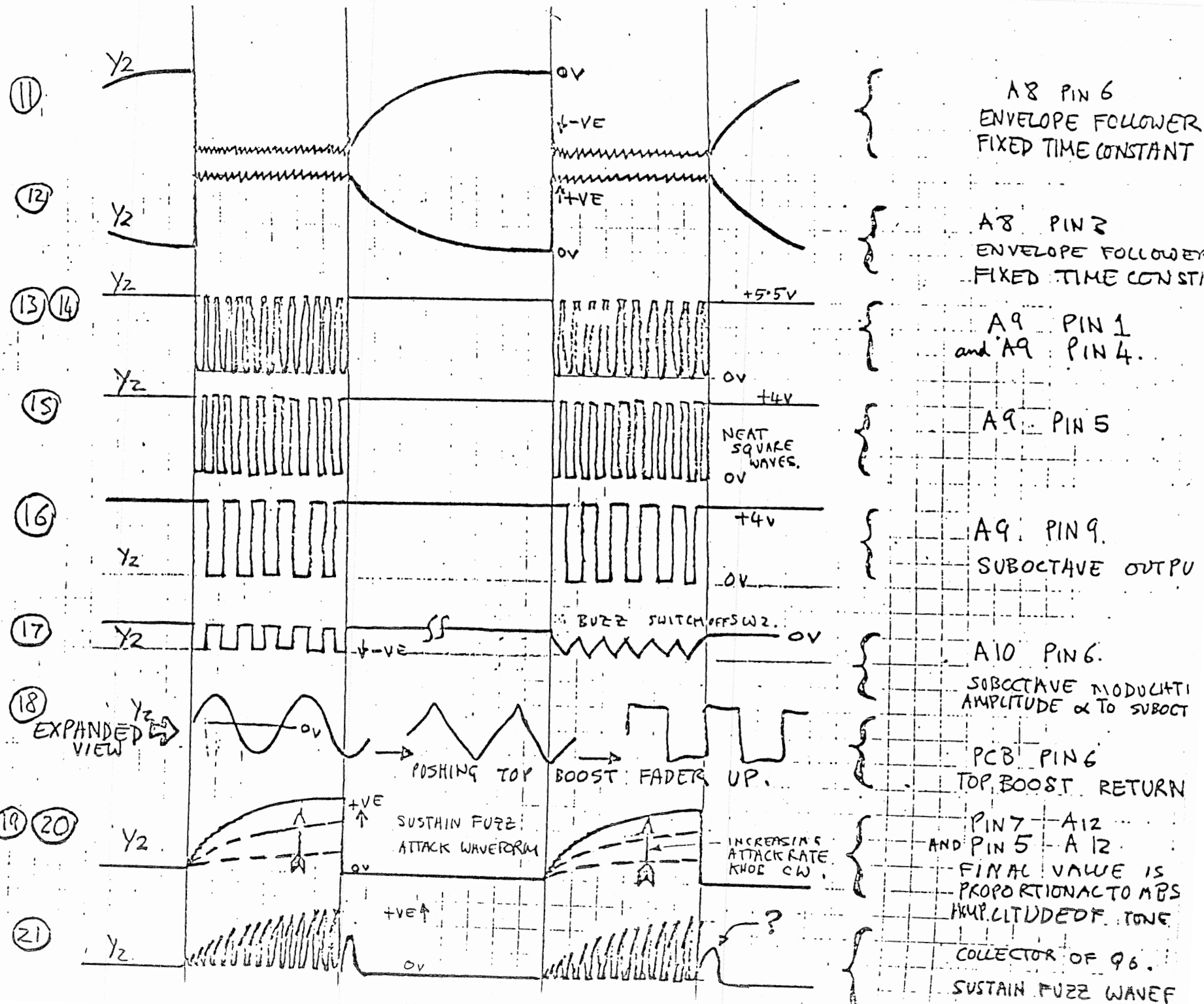
⑧ A5 PIN 1
HYSTERIC RESET PULSE

⑨ A5 PIN 7.
DECAY RATE GENERATOR

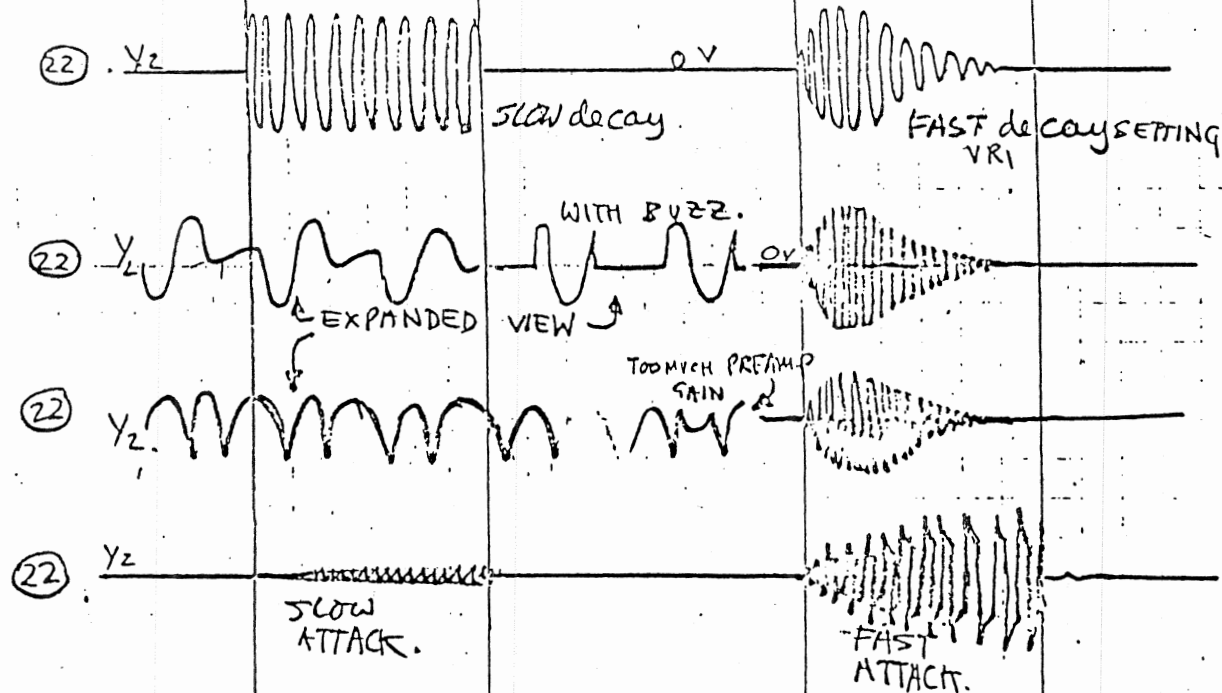
DECAY RATE KNOWN CW.

Test (17). amplitude of signal is controlled by suboctave slider.

Test (19)(20) The attack rate is controlled by the attack rate slider. The final amplitude is proportional to the input signal level. Turn up preamp gain and the waveforms at (19)(20)(21) increase, although the attack time constants remain the same.



WITH the sustain FUZZ
FADER UP AND THE
ATTACK RATE SET
AT SLOW, THE OUTPUT
SHOULD BE VERY SMALL.



AIS PIN 1
All faders down.
output is the unprocessed
signal.

SUBOCTAVE FADER
UP

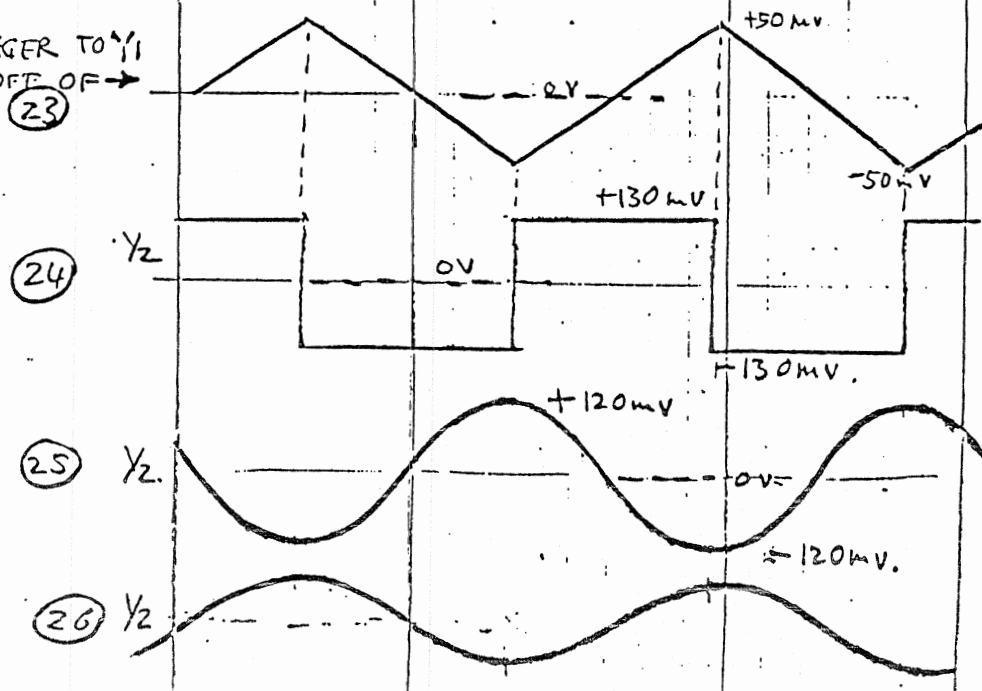
RINGMOD FADER UP

SUSTAIN FUZZ FADER

PHASE CARD.

SWITCH TRIGGER TO $\frac{1}{2}$
TRIGGER OFF OF $\rightarrow$

INTERNAL
FUNCTION
GENERATOR.



AIS PIN 2.
CONTROL MODULATION SWITCH
SET TO $\frac{1}{2}$. FREQUENCY
TRIANGLE CONTROLLED BY
MODULATION SPEED SLIDER
ALSO, TRY $\frac{1}{2}$ MODE

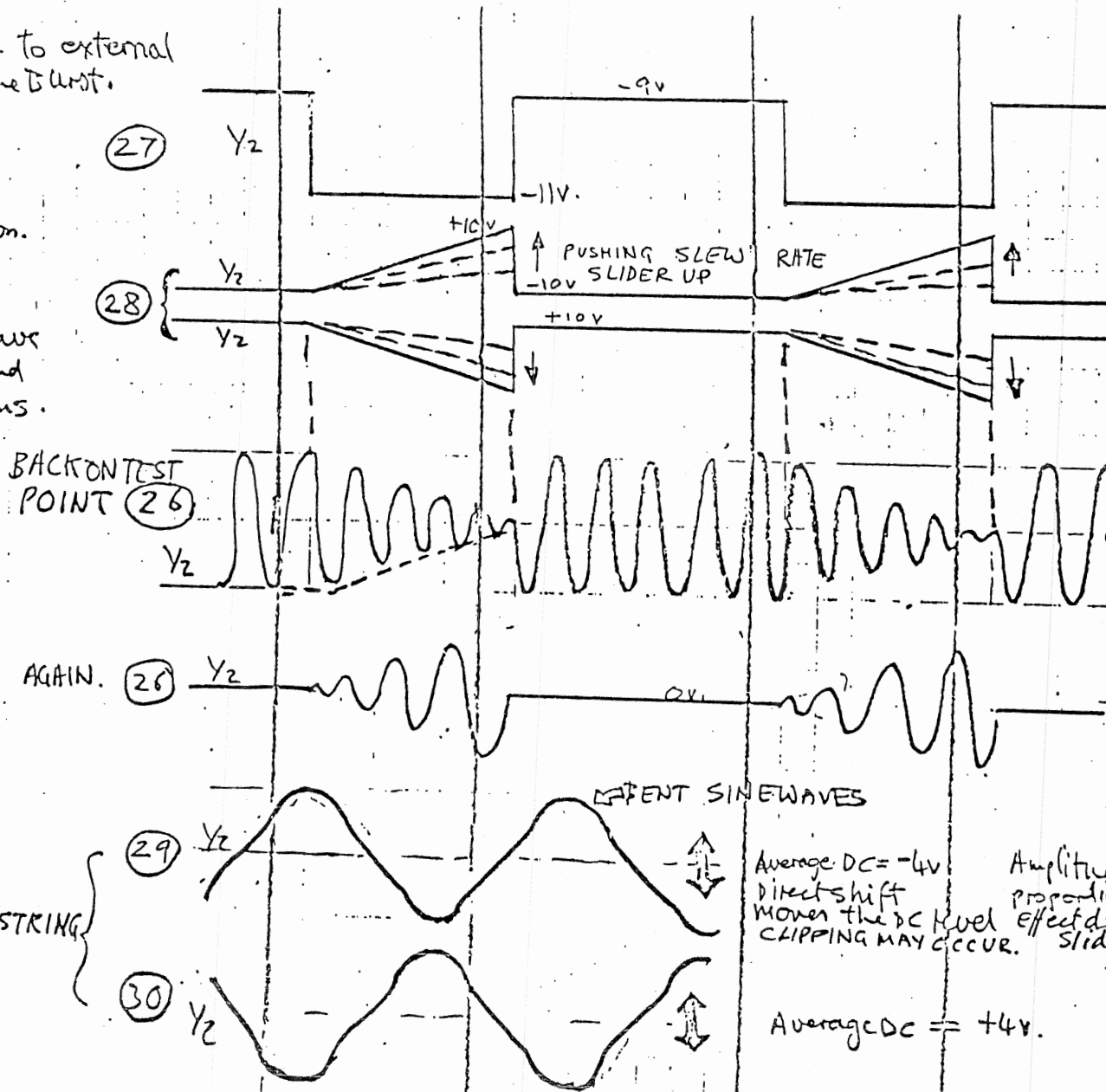
AIS PIN 6.

AIS PIN 6 SNIP THE
SOT RESISTOR IF
WAVEFORM IS ASYMMETRIC.

AIS PIN 6. The Amplitude
should be directly controlled by
effect depth slider. The Direct
slider will directly alter the

SWITCH Back to external trigger. is tone burst.

77.28 is the Ramp modulation. It is combined with the sine wave to give $\sin$ and $\cos$ waveforms.



PCB PIN14.
reset pulse.

A17 PIN6

modulation switch set at $\checkmark$ and at ∇

A12 PIN6.

control modulation set to $\sin$
SWITCH tone burst to slow
CRO Timebase to 10ms/c
EFFECT: Depth to MAXIMUM
DIRECT SHIFT IN CENTRE
MODULATION SPEED to MAX.
SLEW RATE to $\frac{1}{3}$ UP.

AS ABOVE, EXCEPT. control
MODULATION SET TO $\cos$

A10, 7

SWITCH. MODUCATION TO
Modulation Speed slider.
Tone burst set to continuous
Trigger off of Y2

A9, 1 complementary
test point 29 and independent
of control modulation setting
as long as tone burst is set.
continuous.

DIODE BOWSTRING
DRIVERS

SIMILARLY FOR All PIN7 and A8 PIN7 respectively.

* Note that when the modulation effect is set at Meow, the control